

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103**

Winter Semester Examination – December - 2019

Branch: Information Technology

Sem.:- V

Subject: - Probability and Queuing Theory (BTIT504)

Marks: 60

Date:- 16/12/2019

Time:- 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

Q.No	Questions	marks															
Q1	a) Define the terms random variables, variance and standard deviation, moment	04															
	b) State and prove multiplication theorem of probability	04															
	c) Find the probability that in 5 tossing a perfect coin turns up head at least 3 times in succession	04															
Q2	a) What are the conditions under which binomial distribution is used? Mention the equations for distribution function, mean, variance	05															
	b) The hourly wages of 1000 workmen are normally distributed around a mean of Rs 70 and with a standard deviation of Rs 5. Estimate the number of workers whose hourly wages will be i) between Rs 69 and Rs 72. ii) more than 75 iii) less than 63 Or A life insurance salesman sells on the average of 3 life insurance policies per week. Use Poisson's law to calculate the probability that in a given week he will sell 1. Some policies 2. Two or more policies but less than 5 Assuming that there are 5 working days per week, what is the probability that in a given day he will sell one policy	07															
Q3	a) What is Hypothesis? The manufacturer of a patient medicine claimed that it is effective in relieving 90% of the people suffering from the disease. From a sample of 200 people using the medicine 160 were relieved of suffering. Determine if the claim is legitimate	06															
	b) The following data gave the number of air-craft accidents that occurred during the various days of a week. Tests whether the accidents are uniformly distributed over the week <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th>Days:</th><th>Mon</th><th>Tue</th><th>Wed</th><th>Thu</th><th>Fri</th><th>Sat</th><th>Total</th></tr> </thead> <tbody> <tr> <td>No. of accidents:</td><td>15</td><td>19</td><td>13</td><td>12</td><td>16</td><td>15</td><td>90</td></tr> </tbody> </table>	Days:	Mon	Tue	Wed	Thu	Fri	Sat	Total	No. of accidents:	15	19	13	12	16	15	90
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Q4	a)	What are the characteristics of queuing system? Write symbolic representation of queuing model. What do you mean by transient state and steady state queuing systems?	07																				
	b)	In a health clinic, the average rate of arrival of patients is 12 patients per hour. On an average, a doctor can serve patients at the rate of one patient every four minutes. Assume arrival of patients follow Poisson distribution and service to patients follows exponential distribution. i) Find average number of patients in waiting line and in clinic. ii) Find average waiting time in waiting line or in a queue and also average waiting time in clinic.	05																				
Q5	a)	Define Markov process and Stochastic process. Explain transition probability matrix and its Diagrammatic representation	05																				
	b)	Two manufacturers A and B are competing with each other in a restricted market. Over the years, A's customers have exhibited a high degree of loyalty as measured by the fact that customers using A's product 80% of time. Also former customers purchasing the product from B have switched back to A's 60% of time. i) Construct and interpret the state transition matrix in terms of (a) retention and loss (b) retention and Gain ii) Calculate the probability of a customer purchasing A's product at the end of the second period	07																				
Q6	a)	Suppose there are two markets product of brand A and B respectively. Let each of these two brands have exactly 50% of total market in same period and let the market be of a fixed size. The transition matrix given below if the initial market share breakdown is 50% for each brand then determine their market share in the steady state. <table> <tr> <td></td><td></td><td>TO</td><td></td></tr> <tr> <td></td><td></td><td>A</td><td>B</td></tr> <tr> <td></td><td>A</td><td>0.9</td><td>0.1</td></tr> <tr> <td>FRO</td><td>B</td><td>0.5</td><td>0.5</td></tr> <tr> <td>M</td><td></td><td></td><td></td></tr> </table>			TO				A	B		A	0.9	0.1	FRO	B	0.5	0.5	M				06
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	b)	Explain Markov chain and classification of states of Markov chain in details.	06																				

*****End Paper*****